Logarithms

Name _____

Algebra 2 Lesson 6-3 (fast track) Date _____ Period _____

Learning goal: I can write the inverse of an exponential function as a logarithm, and solve an equation by switching between logarithmic form and exponential form.

Language goal: I can explain how I found an inverse or solved an equation using the words base, exponent, argument, inverse, and logarithmic form.

On the quiz: (1) Name the inverse of an exponential function: Example 1, Practice. (2) Estimate ln of a number (full Savvas quiz only). (3) Rewrite a logarithmic equation in exponential form: Example 5 Part B's first step (bonus). (4) Select the logarithms evaluated correctly (full Savvas quiz only). (5) Solve an equation with a common logarithm in it: Example 5.

On the Topic 6 test: Question 7 asks which function is the inverse of an exponential function (Example 1). Question 9 asks you to solve an equation with a base-4 logarithm in it (Example 5).

Exponential and logarithmic forms. A logarithm *is* an exponent.

	Exponential form	**Logarithmic form**
Algebra	$b^x = y$	$\log_b y = x$
Words	the **base** raised to the **exponent** is the result	the log, base b, of the result (the **argument**) is the exponent
Numbers	$3^4 = 81$	$\log_3 81 = 4$

$\log x$ means base 10 (**common logarithm**); $\ln x$ means base e (**natural logarithm**). Spanish: logaritmo.

Example 1: Understand Logarithms

Solve $2^x = 8$. The operation in $2^x = 8$ is exponentiation. To solve this equation, you need an inverse operation that answers the question, "To what exponent would you raise the base 2 to get 8?"

Compare: $2x = 8$. Division is the inverse of multiplication, so you can divide both sides by 2: $\frac{2x}{2} = \frac{8}{2}$, $x = 4$.

Now: $2^x = 8$. A logarithm is the inverse of exponentiation, so you can write $\log_2 8 = x$ (read "log base 2 of 8").
$\log_2 8 = 3$ because $2^3 = 8$.

Definition. $\log_b x = y$ if and only if $b^y = x$, for $b > 0$, $b \neq 1$, and $x > 0$.
The logarithmic function $y = \log_b x$ is the **inverse** of the exponential function $y = b^x$.

Say it: To solve $2^x = 8$, I ask what exponent on the base _____ gives the argument _____. In logarithmic form, $\log_2 8 =$ _____ because _____. The inverse of $y = b^x$ is $y =$ _____.

Try It 1: Understand Logarithms

Write the logarithmic form of $y = 8^x$.

Base _____ exponent _____ result _____ Logarithmic form: _____

Practice: Name the inverse

Write the inverse of each exponential function as a logarithmic function $y = \log_b x$.

a. $y = 5^x$

_____

_____

Inverse: $y =$ _____

b. $y = 10^x$

_____

_____

Inverse: $y =$ _____

c. $y = \left(\frac{1}{2}\right)^x$

_____

_____

Inverse: $y =$ _____

In part b, how do you write the inverse without writing the base? (Look at the box on page 1.)

Example 5: Solve Equations With Logarithms

What is the solution to each equation? Round to the nearest thousandth.

A. $25 = 10^{x-1}$

The variable is in the **exponent**. Convert to logarithmic form (base 10, exponent $x - 1$, result 25):

1. $25 = 10^{x-1}$
2. $\log 25 = x - 1$
3. $1 + \log 25 = x$
4. $2.398 \approx x$

Solution: $x \approx 2.398$.

Common error: 10 is not a coefficient, it is a base. You cannot divide both sides by 10 and then add 1.

B. $\ln(2x + 3) = 4$

The variable is in the **argument**. Convert to exponential form (base e, exponent 4, result $2x + 3$):

1. $\ln(2x + 3) = 4$
2. $2x + 3 = e^4$
3. $2x + 3 \approx 54.598$
4. $2x \approx 51.598$
5. $x \approx 25.799$

Solution: $x \approx 25.799$.

Check: $\ln\Big(2(25.799) + 3\Big) = \ln 54.598 \approx 4$. ✓

Say it: In Part B, I rewrote the logarithmic form $\ln(2x + 3) = 4$ in exponential form. The base is _____, the exponent is _____, and the argument $2x + 3$ equals _____. Then I . . .

Try It 5: Solve Equations With Logarithms

Solve each equation. Round to the nearest thousandth.

a. $\log(3x - 2) = 2$
Base _____ exponent _____ argument _____
Exponential form: _____

$x =$ _____

b. $e^{x+2} = 8$
Base _____ exponent _____ result _____
Logarithmic form: _____

$x \approx$ _____

Why did you switch to exponential form in part a but to logarithmic form in part b? Use the words *exponent* and *argument*.

IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first.

Code	Skill	Feeds
8RK ★	Convert between exponential and logarithmic form: all bases	Quiz 1 and 3; the first step of Quiz 5 (Examples 1 and 5, Practice)
BXU ★	Solve logarithmic equations I	Quiz 5 (Example 5, Try It 5a)
GBR	Evaluate logarithms (optional)	Quiz 4 (full Savvas quiz only)
TDF	Evaluate logarithms using a calculator (optional)	Quiz 2 (full Savvas quiz only); the calculator step in Example 5

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Extension (self-paced): Use Structure

A student says that $\log_3\left(\frac{1}{27}\right)$ simplifies to -3. Is the student correct? Explain.

The student is _____ because the base _____ raised to the exponent _____ gives the argument _____.

Before the quiz, finish these sentences.

A logarithm is an _____: $\log_b x$ is the exponent you put on the base _____ to get the argument _____.

The _____ of the exponential function $y = b^x$ is the logarithmic function $y =$ _____.

When the variable is in the exponent, rewrite the equation in _____ form.

To solve log(expression) $= c$, rewrite it in _____ form: $10^c =$ _____.